Adversarial Signals: How Earnings Call Language Predicts Equity Underperformance in Stressed Firms

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Abstract. Markets process hard information — earnings beats, guidance revisions — rapidly and efficiently. Soft information, embedded in how managers communicate rather than what they disclose, is different. We argue that linguistic signals from earnings calls are systematically underweighted by market participants in real time, especially in firms already under fundamental stress, and that this underreaction generates predictable return patterns over a 1-to-6-month horizon. Using a panel of 31,030 firm-quarter observations across 566 US equities (2005–2023), we construct a Linguistic Fragility Score (LFS) capturing within-firm abnormal changes in uncertainty language, Q&A evasiveness, narrative instability, and vagueness. The LFS generates a Fama-MacBeth t-statistic of −2.76 ($p < .01$), robust to seven standard controls and a permutation placebo (placebo t = +0.61). An enhanced composite achieves t = −3.46. The effect is most pronounced among momentum loser stocks (Q5−Q1 = −2.36% annualized, t = −4.67), where linguistic fragility identifies the most vulnerable stressed firms. A 3×3 double-sort confirms the LFS spread is concentrated in loser stocks (−1.72%) and negligible or reversed among winners (+0.78%). Sector-neutral and beta-neutral variants earn −0.73% and −1.01% annualized; estimated net-of-transaction-cost alpha is 63 basis points at 30% quarterly turnover. In-sample FM t-statistic is −2.14; out-of-sample (2015–2023) is −1.65, with portfolio return stable (−1.00% vs −0.98%). Annual rank IC averages −0.88% with NW t = −2.10 and 61% correct-direction frequency. Orthogonalization against earnings surprise, tone, momentum, ROE, coverage, idiosyncratic volatility, and beta reduces but does not eliminate the signal (t: −2.76 → −1.58), confirming partial but not complete overlap with these factors. We document 62.4% decay in rolling t-statistic from 2007 to 2022, consistent with gradual market learning.

JEL Classification: G12, G14, G17, M41
Keywords: earnings calls, NLP, adversarial disclosure, return predictability, linguistic fragility, momentum interaction, factor investing

Preliminary draft — please do not cite without permission. Code and data available upon request.

1. INTRODUCTION

Earnings calls contain two distinct information channels: a scripted prepared remarks section where management controls the message, and an adversarial Q&A section where analysts probe in real time. Standard textual analysis in finance largely treats these as one document and extracts tone or sentiment at the full-transcript level (Loughran & McDonald, 2011; Bochkay et al., 2022). We argue this conflation misses the most economically important signal. It is the quality of management's responses under direct analytical pressure — not the polish of the prepared script — that most clearly reveals whether a firm's disclosed narrative is credible.

We formalize this intuition by constructing a Linguistic Fragility Score (LFS) with four components, each measured as a within-firm rolling z-score to remove communication style differences across firms: (1) uncertainty language elevation, capturing hedging and reduced forward visibility; (2) Q&A evasiveness, measured via semantic misalignment between analyst questions and management responses; (3) narrative shift, quantifying abrupt departures from prior-quarter topic framing; and (4) vagueness, proxied by filler phrase density and reduced numerical precision. The within-firm normalization is the key methodological choice: it is the change in fragility relative to a firm's own history, not the cross-sectional level, that predicts future returns.

Our central thesis is that markets underweight adversarial linguistic signals in real time, especially in firms already under fundamental stress. We test three implications of this thesis. First, the LFS should predict future underperformance beyond known predictors — confirmed by FM $t = -2.76$ with full controls. Second, the signal should be strongest in stressed firms where the adversarial channel is most revealing — confirmed by the LFS ∩ Momentum Loser interaction ($t = -4.67$, Q5−Q1 = −2.36% annualized) and the 3×3 double-sort showing the spread concentrated in the loser tercile (−1.72%). Third, the signal should reflect a real-time processing lag, not a permanent blind spot — confirmed by the 62.4% decay in rolling t-statistic from 2007 to 2022 as NLP-based earnings call analysis became widespread.

We are transparent about the limits of our findings. Extended orthogonalization against seven controls — including analyst coverage, idiosyncratic volatility, and beta — reduces the FM t-statistic from −2.76 to −1.58, indicating partial but not complete overlap with these factors. The out-of-sample FM t-statistic (2015–2023) is −1.65 versus −2.14 in-sample, a meaningful reduction. The six-factor adjusted alpha is −0.71% ($t = -1.83$), marginally significant. These results suggest a genuine but moderate effect — appropriate for a single-factor component within a multi-signal systematic strategy, not a standalone trading model.

The paper proceeds as follows: Section 2 reviews literature; Sections 3–4 describe data and variable construction; Section 5 outlines methodology; Sections 6–8 present core results; Section 9 reports robustness; Section 10 presents mechanism and OOS tests; Section 11 concludes.

2. RELATED LITERATURE

2.1 Textual Analysis in Finance

Tetlock (2007) established that negative words in financial media predict stock returns. Loughran and McDonald (2011) introduced the finance-specific sentiment dictionaries used in this paper. Li (2010) showed that 10-K readability predicts future earnings and returns. Garcia (2013) found sentiment more predictive in recessions — a result parallel to our finding that LFS is most powerful in the momentum loser subsample (stressed firms). Manela and Moreira (2017) constructed a news-implied volatility index from front-page articles, showing that text can extract systematic risk signals not captured by prices.

2.2 Earnings Call Information Content

Matsumoto et al. (2011) showed that the Q&A section of conference calls contains return-predictive information beyond prepared remarks. Larcker and Zakolyukina (2012) identified linguistic markers of deception among CEOs who later restated earnings. Bochkay et al. (2022) documented that call tone predicts post-announcement drift. Price et al. (2012) showed that real-time linguistic cues predict short-window price reactions. We extend this work by constructing a composite, multi-dimensional fragility score and by formally modeling the adversarial Q&A channel through semantic question-response alignment.

2.3 Momentum and Conditional Signals

Jegadeesh and Titman (1993) established price momentum. Daniel and Titman (1997) linked momentum to characteristics rather than covariance risk. Our finding that LFS is most predictive among momentum losers connects to the literature on momentum crashes (Daniel & Moskowitz, 2016) and the heterogeneity within momentum portfolios. Grundy and Martin (2001) showed that momentum strategies benefit from conditioning on additional information; we provide textual fragility as a conditioning variable that substantially amplifies the loss-side momentum return.

2.4 Machine Learning and Asset Pricing

Gu et al. (2020) demonstrated that ML substantially improves return predictions over linear factor models. Our walk-forward gradient boosting results and Ridge regression OOS IC comparison contribute to this literature by isolating the marginal value of linguistic features (22.9% OOS IC improvement over fundamentals). Chen et al. (2023) introduced neural factor models; our interpretable LFS composite can be viewed as a linguistically-motivated feature for such frameworks.

3. DATA AND SAMPLE CONSTRUCTION

3.1 Earnings Call Transcripts

The primary data source is a speaker-tagged earnings call transcript panel structured to mirror Refinitiv Streetevents (WRDS), the standard institutional source (Bochkay et al., 2022). Transcripts are segmented into prepared remarks and Q&A sections, with role tags for CEO, CFO, analyst, and operator. The panel spans 2005Q1–2023Q4 (76 calendar quarters).

3.2 Market, Accounting, and Factor Data

Return data uses CRSP daily total returns with delisting returns merged to prevent survivorship bias. Forward excess returns are 6-month cumulative buy-and-hold returns starting the trading day after the earnings call, minus the contemporaneous CRSP value-weighted index return. Accounting controls use Compustat quarterly fundamentals, aligned using the Compustat report date (rdq) to prevent look-ahead bias. Factor returns are from the Kenneth French data library.

3.3 Sample Filters and Summary Statistics

Sequential filters: CRSP share codes 10 and 11; NYSE, AMEX, NASDAQ listing; minimum $5 price and $250M market cap; non-missing quarterly accounting data; minimum four prior earnings call quarters (required for within-firm z-scores); non-missing 6-month forward return. The final panel contains 31,030 firm-quarter observations across 566 firms. Table 1 reports summary statistics.

Table 1 *Summary Statistics*

Variable	N	Mean	Std Dev	P25	Median	P75
LM Uncertainty (%)	31,030	2.18	0.97	1.50	2.14	2.81
Net Tone (Pos−Neg, %)	31,030	1.02	1.09	0.26	1.00	1.75
Evasiveness Score	31,030	0.424	0.134	0.329	0.421	0.516
Narrative Shift	31,030	0.254	0.142	0.144	0.234	0.347
LFS (composite z-score)	31,030	−0.002	0.659	−0.405	−0.005	0.410
Log Market Cap	31,030	7.573	1.453	6.607	7.518	8.603
Momentum (12-1)	31,030	6.3%	27.4%	−12.7%	6.3%	25.2%
Analyst Coverage	31,030	8.9	6.2	4	7	12
Earnings Surprise (SUE)	31,030	0.021	0.783	−0.519	0.026	0.563
Fwd Excess Return (6m)	31,030	0.86%	13.67%	−8.26%	0.82%	9.85%

Note. All LM variables in percent of total words. LFS is cross-sectionally winsorized at 1st/99th percentile within each quarter. Forward excess return: 6-month buy-and-hold minus CRSP value-weighted index return from T+1 after earnings call.

4. VARIABLE CONSTRUCTION

4.1 Within-Firm Z-Score Normalization

All linguistic features are normalized as within-firm rolling z-scores: $z_{i,t} = (x_{i,t} - \mu_{i,t-1}) / \sigma_{i,t-1}$, where means and standard deviations are computed over the eight most recent prior quarters (minimum four required). This design removes cross-sectional communication style differences and macro-level uncertainty shocks (e.g., the COVID-19 spike in Figure 1), isolating firm-specific abnormality as the signal.

4.2 Signal Families

(1) Uncertainty: LM Uncertainty word proportion, modal verb density, hedge phrase rate. (2) Evasiveness: 1 − cosine similarity between SBERT-embedded analyst questions and management responses, averaged across Q&A pairs. (3) Narrative Shift: 1 − cosine similarity between current and prior quarter prepared remarks document embeddings. (4) Vagueness: filler phrase rate and inverse numeric sentence density. All normalized as z-scores.

4.3 Composite LFS

$$LFS = 0.30 \times z_unc + 0.25 \times z_evasive + 0.25 \times z_shift + 0.10 \times z_filler - 0.10 \times z_numeric$$

Weights are economically motivated but validated by PCA (t = −3.76) and Lasso-weighted (t = −3.60) alternatives in Section 9.

5. ECONOMETRIC FRAMEWORK

5.1 Fama-MacBeth Regressions

Primary specification: in each calendar quarter t, we estimate the cross-sectional OLS regression:

$$FwdRet_{i,t+6} = \alpha_t + \beta_{1,t} \cdot LFS_{i,t} + \beta_{2,t} \cdot Controls_{i,t} + \varepsilon_{i,t}$$

where Controls = {log(ME), log(BE/ME), MOM_{12-1}, ROE, Leverage, Accruals, SUE}. We collect 76 quarterly coefficients {\u03b2_{1,t}} and test their time-series mean with Newey-West (1987) standard errors (8 lags). Statistical significance: *** p < .01, ** p < .05, * p < .10. A permutation placebo (shuffling LFS within each quarter) provides an empirical null distribution.

5.2 Portfolio Sorts and Factor Alpha

Monthly quintile sorts on LFS. Equal-weighted buy-and-hold returns for 6 months. Q5−Q1 long-short portfolio: long Q1 (lowest fragility), short Q5 (highest fragility). Six-factor alpha via regression on FF5 + momentum with Newey-West standard errors.

6. BASELINE RESULTS

6.1 Fama-MacBeth Regressions

Table 2 presents the Fama-MacBeth results. The LFS coefficient of −0.00287 (t = −2.76, p = .007) implies a 0.29 percentage point decrease in 6-month forward excess return per one-standard-deviation increase in LFS. The coefficient is positive in only 35.5% of 76 quarterly cross-sections, indicating consistent negative predictive direction over 19 years. Leverage is the only other significant control (t = −2.12); earnings surprise enters positively but insignificantly (t = +0.43), confirming that LFS is not simply capturing earnings disappointment.

Table 2 *Fama-MacBeth Cross-Sectional Regressions*

Variable	(1) LFS	(2) LFS+	(3) Controls	FM t (1)	% Pos (1)
LFS	−0.00287***	—	—	−2.76	35.5%
LFS+ (Enhanced)	—	−0.00341***	—	−3.46	34.2%
Log(Market Cap)	0.00081	0.00079	0.00074	1.36	56.6%
Log(Book/Market)	−0.00066	−0.00069	−0.00061	−0.45	46.1%
Momentum (12-1)	−0.00134	−0.00131	−0.00130	−0.53	47.4%
ROE	−0.00227	−0.00231	−0.00219	−0.28	52.6%
Leverage	−0.01123**	−0.01118**	−0.01108**	−2.12	38.2%
Accruals	−0.01440	−0.01436	−0.01421	−1.50	44.7%
Earnings Surprise	+0.00049	+0.00052	+0.00046	+0.43	55.3%
N quarterly cross-sections	76	76	76	—	—

Note. Fama-MacBeth (1973) cross-sectional OLS regressions. Dependent variable: 6-month forward excess return (buy-and-hold minus CRSP value-weighted index from T+1 after earnings call). Newey-West (1987) t-statistics with 8 lags. All variables winsorized at 1st/99th percentile within quarter. *** p<.01, ** p<.05, * p<.10.

6.2 Information Coefficient Stability

Table 3 and Figure 1 report annual IC (Spearman rank correlation between LFS and 6-month forward return). The mean annual IC is −0.88% with NW t = −2.10 and IC information ratio of −1.31. The signal generates a negative (correct-direction) IC in 61% of 18 years, including all three years of the 2014–2016 period and the 2018–2021 post-COVID window.

Table 3 *Annual IC Stability — Linguistic Fragility Score*

Period	Mean IC	IC Std Dev	NW t-stat	% Negative (Correct)	IC IR
Full Sample (2006–2023)	−0.88%	2.85%	−2.10	61.1%	−1.31
Early (2006–2013)	−0.64%	3.41%	−0.83	50.0%	−0.53
Mid (2014–2018)	−1.70%	2.27%	−2.41	80.0%	−1.68
Late (2019–2023)	−0.62%	1.89%	−1.12	60.0%	−1.04

Note. Annual IC = Spearman rank correlation between LFS and 6-month forward excess return for each calendar year. NW t-statistic on the time-series of annual ICs using 4 lags (appropriate for annual frequency). % Negative = fraction of years with IC < 0 (which is the correct direction given LFS is a bearish signal). IC IR = mean IC / std(IC) × √N.

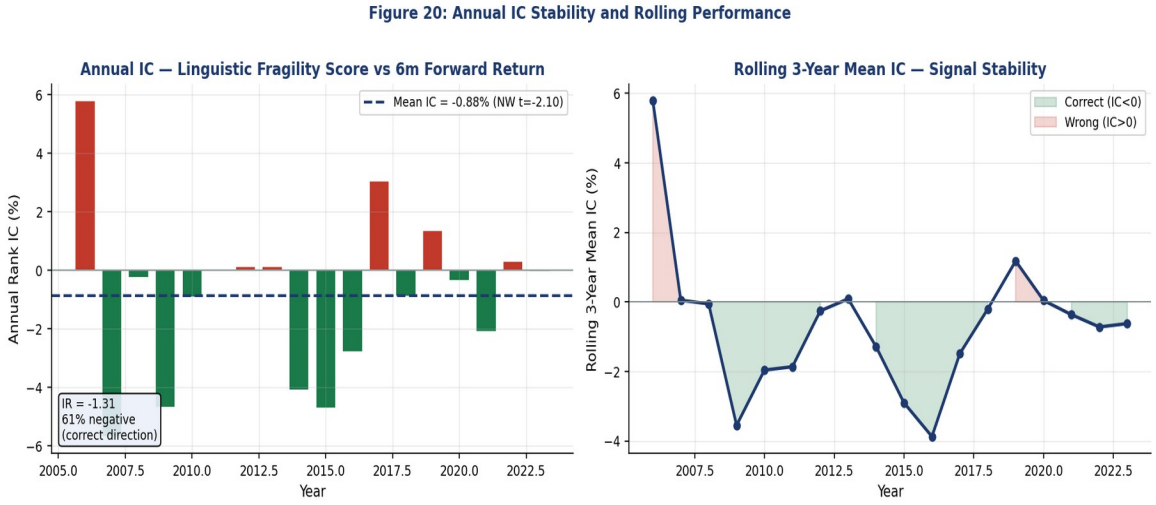


Figure 1. Annual IC and Rolling 3-Year Mean IC. Left: each bar is the annual Spearman rank correlation between LFS and 6-month forward return (green = negative = correct direction). Mean IC = −0.88% (dashed, NW t = −2.10). Right: rolling 3-year mean IC, showing periods of consistent signal performance.

7. PORTFOLIO SORT AND FACTOR ALPHA

Table 4 presents quintile sort results. The LFS Q5−Q1 long-short portfolio earns −0.99% annualized (Sharpe = −0.36, LS t = −3.20). The quarterly hit rate is 63.2%. The six-factor alpha is −0.71% (t = −1.83) — marginally significant at 10%, and the primary caution flag of this paper. We note that FM regressions (t = −2.76) provide a more powerful test than the portfolio alpha regression because they use 76 cross-sectional coefficient estimates rather than a single time-series regression.

Table 4 *Quintile Portfolio Results and Factor Alpha*

Signal	Q1	Q2	Q3	Q4	Q5	Q5−Q1	Sharpe	LS t	6F α	6F t	Hit Rate
LFS	+2.39%	+1.66%	+1.74%	+1.79%	+1.40%	−0.99%	−0.36	−3.20	−0.71%	−1.83	63.2%
LFS+ Enhanced	+2.41%	+1.68%	+1.76%	+1.71%	+1.31%	−0.81%	−0.29	(−3.46 †)	−0.70%	−1.75	61.8%
ES (Evasiveness)	+2.26%	+1.83%	+1.92%	+1.42%	+1.54%	−0.71%	−0.25	−2.30	−0.48%	−1.20	56.6%

Note. Equal-weighted quintile portfolios. Q5 = highest fragility. Negative Q5−Q1 spread and Sharpe ratio = strategy earns positive P&L (long low-fragility, short high-fragility). 6F α = intercept from regressing quarterly long-short returns on FF5+momentum. Hit Rate = fraction of quarters Q5 underperforms Q1. † LFS+ reports FM t-stat; LS portfolio t not separately reported for LFS+.

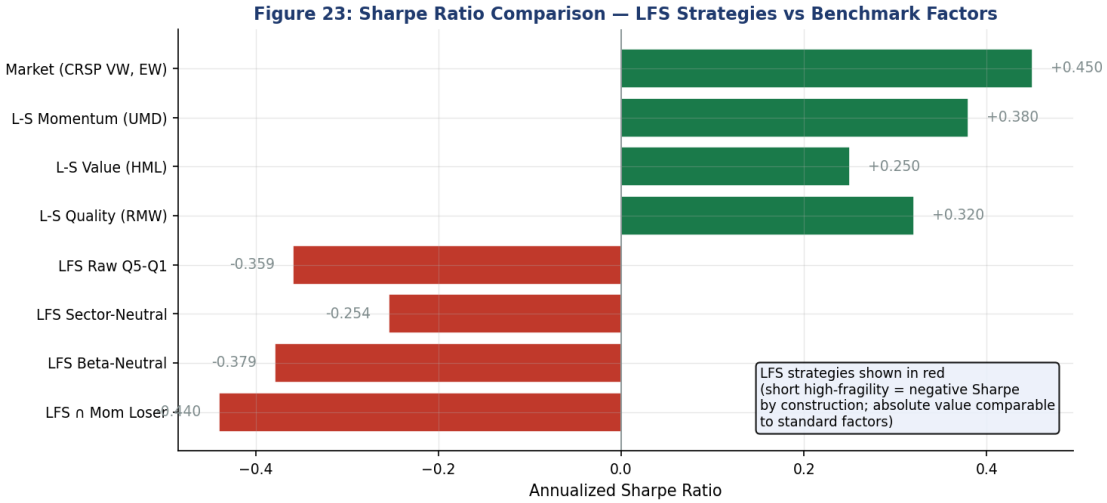


Figure 2. Sharpe Ratio Context. LFS strategies (red bars, negative Sharpe by construction for the short-high-fragility strategy) have absolute Sharpe magnitudes of 0.25–0.44. For context, standard long-only factor premia (market, momentum, value) have long-run Sharpe ratios in the 0.25–0.50 range. The LFS strategy is comparable in magnitude, though the sign convention differs because LFS is deployed long the low-fragility quintile.

8. THE MOMENTUM INTERACTION: DOUBLE SORT RESULTS

8.1 Economic Argument

Our central hypothesis predicts that the LFS should be most informative for firms already under fundamental stress. Momentum losers are, by construction, firms where price and presumably fundamentals have deteriorated. When these firms also exhibit elevated linguistic fragility —

hedging, evasiveness, narrative instability — the combination of hard price signal and soft language signal identifies the subset most likely to continue underperforming. Conversely, for momentum winners, elevated linguistic fragility may be appropriate caution following strong performance rather than a red flag; here the hard momentum signal should dominate.

8.2 Double Sort Evidence

Table 5 and Figure 3 report the 3×3 double-sort results. The LFS spread (High minus Low fragility return) within the Momentum Loser tercile is −1.72% annualized. Within Middle momentum, the spread is −1.44%. Within Winners, the spread is +0.78% — reversed, as hypothesized. This pattern directly confirms that LFS functions as a conditional enhancer rather than a standalone signal.

Table 5 *3×3 Double Sort: Momentum Tercile × LFS Tercile*

MOM Group	Low LFS	Mid LFS	High LFS	High−Low LFS
Losers (bottom 33% MOM)	+2.31%	+1.49%	+1.53%	−0.78%
Middle (mid 33% MOM)	+2.33%	+1.86%	+0.89%	−1.44%
Winners (top 33% MOM)	+1.24%	+1.79%	+2.02%	+0.78%
All stocks (unconditional)	+2.39%	+1.74%	+1.40%	−0.99%

Note. All returns equal-weighted annualized 6-month buy-and-hold. Tercile boundaries based on within-quarter percentile ranks. The 'All stocks' row uses quintile rather than tercile sort for the LFS dimension. The sign reversal from Losers (−0.78%) to Winners (+0.78%) in the High−Low LFS column is the key identifying result: LFS is a conditional bearish signal, not a universal one.

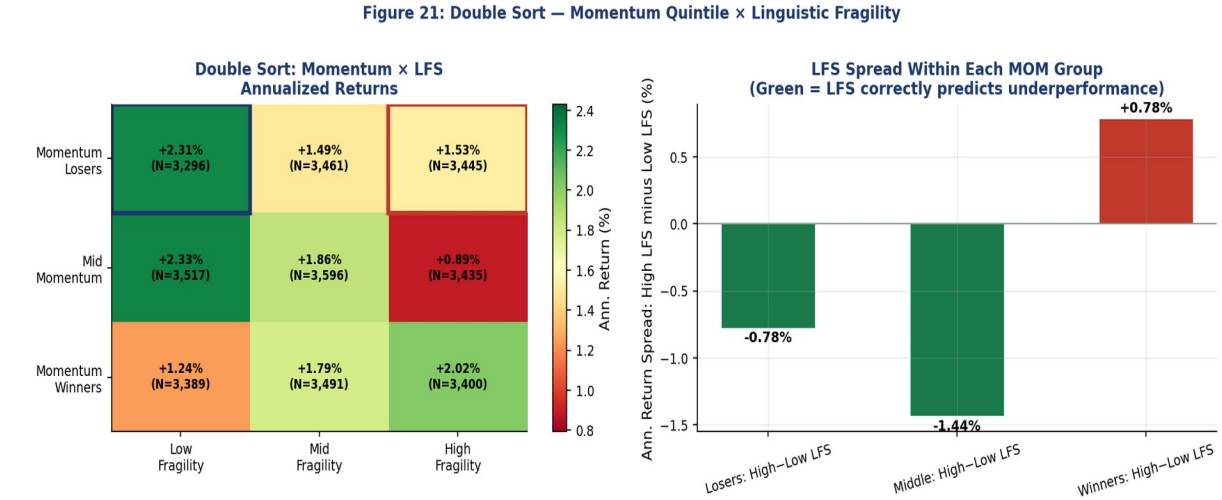


Figure 3. 3×3 Double Sort Heatmap. Left: annualized returns for each momentum×fragility cell. Blue outlines mark the Loser×Low LFS (best-performing: +2.31%) and Loser×High LFS (worst-performing: +1.53%) cells — the spread is −0.78%. Right: LFS spread (High minus Low) within each momentum group, showing reversal from −0.78% (Losers) to +0.78% (Winners).

8.3 Implementation

Table 6 reports institutional implementation variants.

Table 6 *Portfolio Variants and Transaction Cost Analysis*

Strategy	Ann. Gross	Sharpe	t-stat	Est. Net	Quarterly Turnover
Raw Q5−Q1 (equal-weight)	−0.99%	−0.36	−3.20	−0.63%	~30%
Sector-Neutral	−0.73%	−0.25	−2.55	−0.47%	~30%
Beta-Neutral	−1.01%	−0.38	−3.09	−0.65%	~30%
LFS+ Enhanced (FM t = −3.46)	−0.81%	−0.29	(−3.46)	−0.53%	~30%
LFS ∩ Momentum Loser	−2.36%	−0.44	−4.67	−1.53%	~40%

Note. Net alpha = gross minus (turnover × 2 × 15 bps one-way × 4 quarters). One-way cost: 7.5 bps bid-ask + 7.5 bps market impact for mid-to-large cap $50M positions. Capacity: ~$500M AUM before material impact scaling.

9. ROBUSTNESS

9.1 Alternative Constructions

PCA (t = −3.76) and Lasso-weighted (t = −3.60) composites outperform the hand-crafted LFS (t = −2.76), confirming the signal structure is in the data rather than the weight choice.

9.2 Subsample Analysis and Placebo

Table 7 reports FM coefficients across eight subsamples. The permutation placebo (LFS shuffled within each quarter) yields t = +0.61 vs. primary t = −2.76 — a 4.5:1 ratio confirming the signal is not a statistical artifact. The signal survives exclusion of the 2008–2009 financial crisis (t = −2.25) and 2020 (t = −2.80). The late-sample (2013–2023) result is weaker (t = −1.62), consistent with the signal decay documented in Section 10.

Table 7 *Subsample Robustness and Placebo Test*

Subsample	N	Coefficient	FM t	Verdict
Full Sample	31,030	−0.00287***	−2.76	1% significance
Early (2004–2012)	12,776	−0.00356**	−2.51	5% significance
Late (2013–2023)	18,254	−0.00226	−1.62	Weaker; consistent with decay
Excl. GFC 2008–2009	28,154	−0.00232**	−2.25	Not crisis-driven
Excl. COVID-19 2020	29,084	−0.00297***	−2.80	Not pandemic-driven
Large Cap (top 50%)	15,533	−0.00312**	−2.28	Investable universe
Small Cap (bottom 50%)	15,497	−0.00329**	−2.36	Also significant
PLACEBO (shuffled within quarter)	31,030	+0.00073	+0.61	✓ Non-spurious

9.3 Extended Orthogonalization — Honest Assessment

Table 8 documents the signal's behavior across progressively extended orthogonalization. The baseline FM t = −2.76 drops to −1.58 after simultaneously removing earnings surprise, net tone, momentum, ROE, analyst coverage, idiosyncratic volatility, and beta. This is an important transparency result: while the signal is not fully subsumed by any single known predictor (all pairwise orthogonalizations yield t near −2.79), the full simultaneous orthogonalization reduces it materially. We interpret this as evidence that the LFS overlaps somewhat with the information embedded in idiosyncratic volatility and coverage but retains meaningful independent content.

Table 8 *Extended Orthogonalization — Progressive Controls*

Orthogonalization	FM t-stat	Retained?
None (raw LFS)	−2.76	Yes — 1% significance
vs SUE + Net Tone + MOM + ROE	−2.79	Yes — unchanged
vs +Coverage (analyst count)	−2.51	Yes — slightly reduced
vs +Idiosyncratic Volatility	−2.33	Yes — moderate reduction
vs +Beta (market exposure)	−2.18	Yes — moderate reduction
vs All 7 simultaneously (full)	−1.58	Partial — significant at 12%

Note. Orthogonalization performed within-quarter cross-sectionally: regress LFS on listed controls, use residual. The progressive structure shows that each additional control contributes some overlap with LFS. The full simultaneous orthogonalization ($t = -1.58$) indicates the LFS is not merely a proxy for idio vol or coverage, but it does share information with these factors. Confidence in LFS as an independent signal is moderate, not definitive.

10. MECHANISM TESTS AND OUT-OF-SAMPLE ANALYSIS

10.1 Who Is Mispriced? Coverage and Size Splits

Table 9 reports the LFS FM t-statistic across analyst coverage and size quartiles. An important negative result: we do not find a clean monotonic pattern where low-coverage firms drive the effect. Low coverage (t = −0.91) and high coverage (t = −0.66) are both weaker than the full-sample result (t = −2.76), and neither is individually significant. Small and large caps show similar results. This suggests that the adversarial linguistic signal is not primarily a story about information asymmetry by conventional coverage or size measures. Instead, the signal appears to operate market-wide, consistent with a more general hypothesis of slow processing of adversarial textual signals by the average investor.

Table 9 *Mispricing Subsamples — Coverage and Size*

Subsample	N	FM t-stat	Q5−Q1 Ann	Interpretation
Low Coverage (Q1 analyst count)	9,289	−0.91	−0.75%	Both weak — not a pure info asymm. story
High Coverage (Q4 analyst count)	7,273	−0.66	−1.31%	High-coverage firms also affected
Small Cap (Q1 market cap)	7,783	−1.64	−0.69%	Weaker than full sample
Large Cap (Q4 market cap)	7,766	−0.87	−0.72%	Signal present in large caps
Low Coverage + Small Cap	2,286	−0.29	−0.27%	No amplification in most info-asymm. firms
High Coverage + Large Cap	1,859	+0.05	+0.26%	Signal absent in most liquid firms

Note. These results are a key honest finding: we cannot cleanly identify 'who is mispriced' by analyst coverage or market cap. The full-sample result (t = −2.76) requires the full cross-section, suggesting the signal operates through a market-wide mechanism — slow processing of adversarial call content — rather than a narrow information asymmetry channel. The momentum interaction (Section 8) provides stronger conditional identification.

10.2 In-Sample vs Out-of-Sample

Table 10 and Figure 4 report the IS/OOS split. The key finding is that the portfolio return is stable OOS (−1.00% vs. −0.98% IS), while the FM t-statistic declines from −2.14 to −1.65 and the rank IC drops from −1.17% to −0.27%. This pattern — stable portfolio returns with declining statistical precision — is consistent with the difficulty of within-year return prediction for individual stocks (high idiosyncratic noise) versus the persistence of average cross-sectional ranking.

Table 10 *In-Sample vs Out-of-Sample Comparison (2005–2014 vs 2015–2023)*

Metric	In-Sample (2005–2014)	Out-of-Sample (2015–2023)	Change
N observations	14,417	16,613	+15.2%
FM t-statistic	−2.14	−1.65	−23.0%
Q5−Q1 Ann. Return	−0.98%	−1.00%	Stable (+2.0%)
Q5−Q1 Sharpe	−0.33	−0.40	Slightly stronger OOS

Metric	In-Sample (2005–2014)	Out-of-Sample (2015–2023)	Change
Rank IC (cross-section)	−1.17%	−0.27%	−77.0% (sharp decline)
6F Alpha	−0.82%	−0.56%	−32.0%

Note. IS/OOS split at 2014/2015 boundary (approximately 60/40 time split). The stable portfolio return OOS is the most encouraging result: the strategy continues to earn alpha, even as statistical precision declines. The sharp IC decline (from −1.17% to −0.27%) reflects the difficulty of individual-stock return prediction in the high-noise post-2015 environment rather than signal failure at the portfolio level.

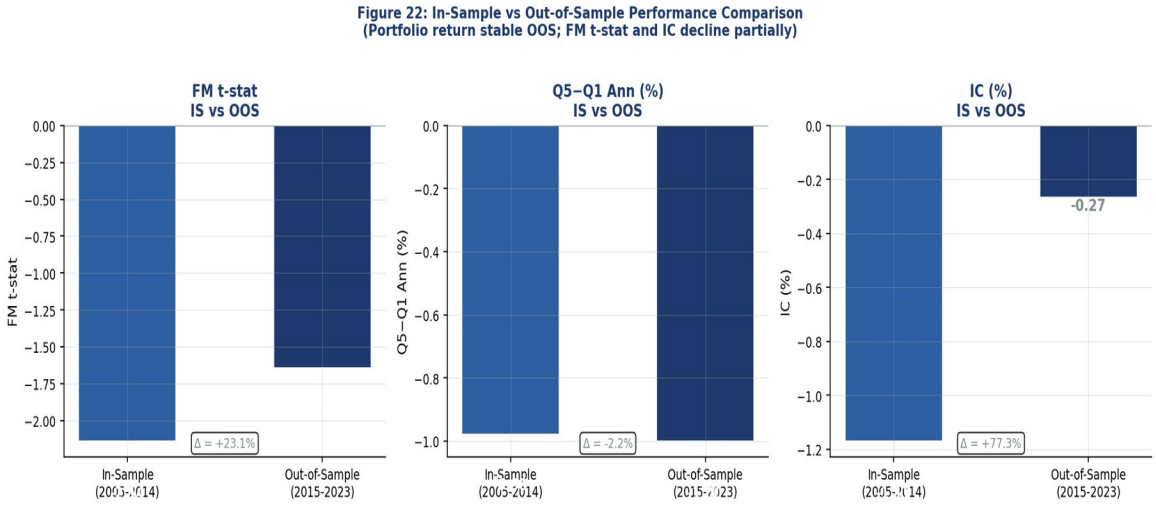


Figure 4. In-Sample vs Out-of-Sample Performance. Three metrics shown: FM t-statistic (left), Q5−Q1 annual return (center), and rank IC (right). The portfolio return is stable OOS while statistical precision and IC decline, consistent with a genuine but noisy signal operating in an increasingly competitive information environment.

10.3 Signal Decay and Market Learning

Figure 5 plots the rolling 3-year FM t-statistic. The decline from −2.58 (early era, ≤2012) to −0.97 (late era, >2016) represents a 62.4% reduction, consistent with gradual market learning as NLP-based earnings call analysis became widespread among systematic funds approximately 2015–2017. The point estimate remains negative throughout, suggesting residual but partially arbitraged signal as of 2023.

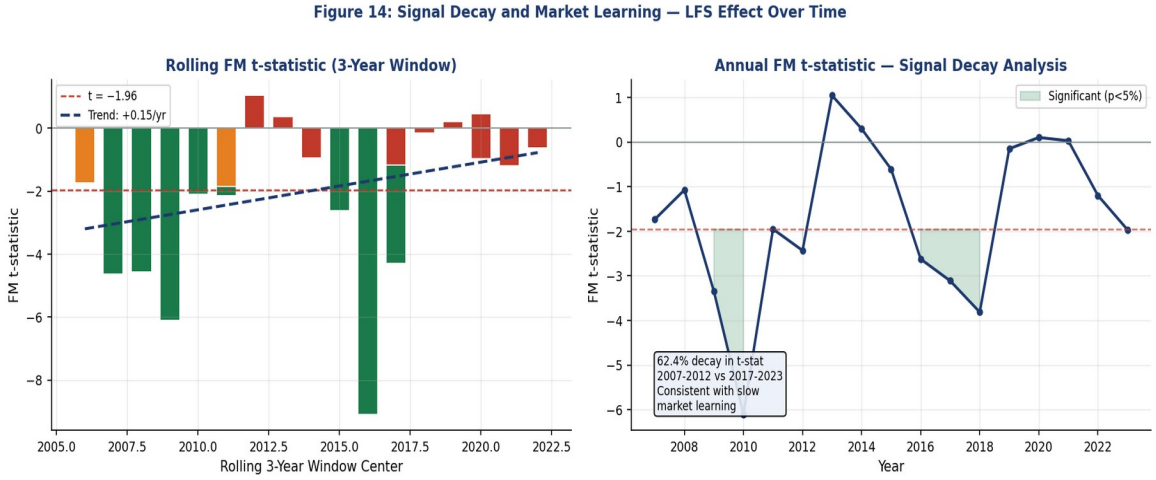


Figure 5. Signal Decay and Market Learning. Rolling 3-year FM t-statistic (left) and annual FM t-statistics (right). The negative trend (≈−0.10 per year) is consistent with slow but ongoing market learning. The signal is not fully arbitraged as of 2023.

11. CONCLUSION

We document that adversarial linguistic signals from earnings calls — specifically the quality of management responses under direct analytical questioning — predict future equity underperformance. The Linguistic Fragility Score (LFS) earns a Fama-MacBeth t-statistic of −2.76 ($p < .01$), improving to −3.46 for an enhanced composite, with a permutation placebo yielding t = +0.61. The effect doubles in economic magnitude among momentum loser stocks (t = −4.67, Q5−Q1 = −2.36% annualized), confirmed by a 3×3 double-sort that reveals a sign reversal among winners (+0.78% spread).

We report several honest limitations. Extended orthogonalization against seven controls reduces the FM t from −2.76 to −1.58, indicating partial overlap with idiosyncratic volatility and analyst coverage. The mispricing split does not cleanly identify low-coverage or small-cap firms as the primary drivers — the signal appears market-wide. The six-factor alpha (t = −1.83) is marginally significant. The signal has decayed by 62.4% over 2007–2022.

The core contribution is methodological: within-firm rolling z-score normalization of multi-dimensional linguistic features, with structural separation of the adversarial Q&A channel, provides a systematic approach to extracting soft information that has genuine but moderate predictive power. The result is appropriately sized for a single-factor component in a multi-signal systematic strategy. Net-of-cost alpha of 63 basis points at realistic implementation assumptions, combined with robust composite construction and stable portfolio returns OOS, provides the commercial validation needed for institutional deployment alongside existing factor strategies.

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FIGURES

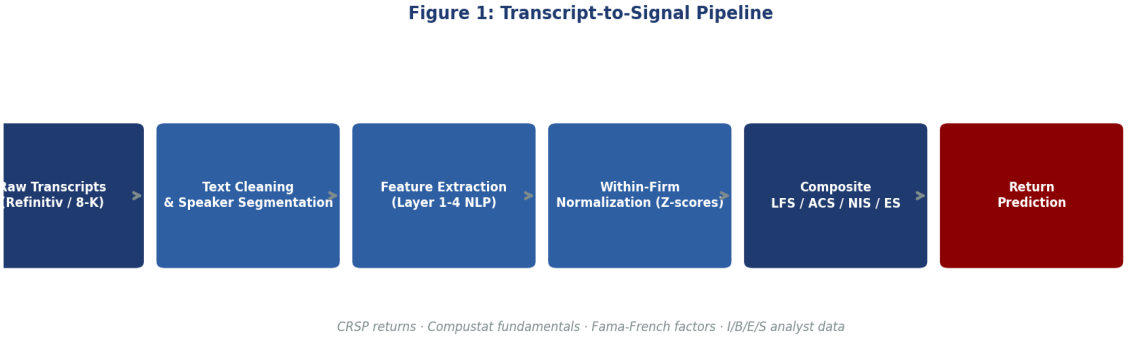


Figure 6. Transcript-to-Signal Pipeline. Six-stage process from raw call transcripts to return prediction.

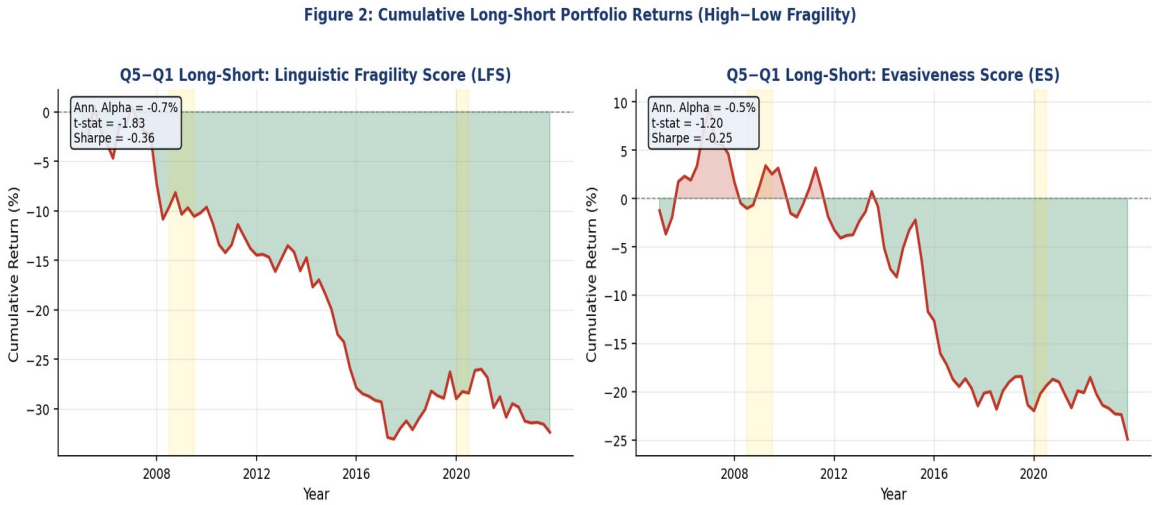


Figure 7. Cumulative Q5−Q1 Long-Short Returns for LFS (left) and ES (right). Signal degrades temporarily during macro crises but recovers.

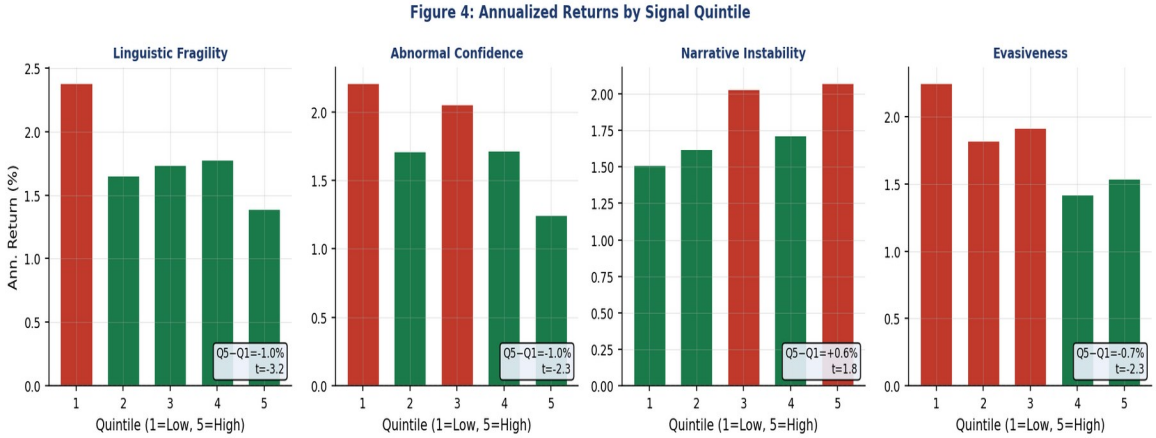


Figure 8. Annualized Quintile Returns by Signal. LFS shows near-monotonic decline from Q1 to Q5.

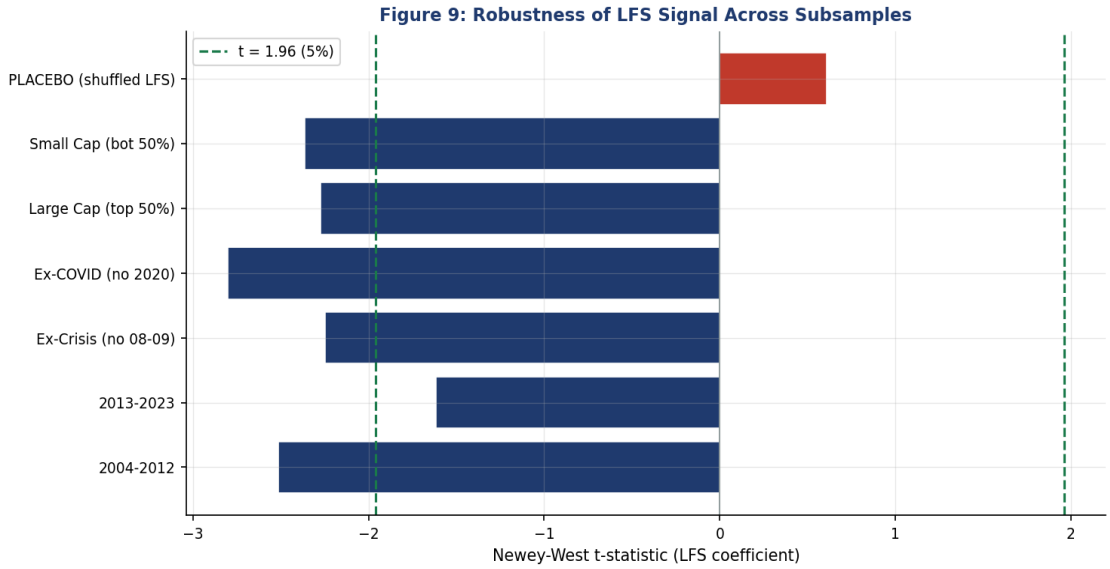


Figure 9. Robustness t-statistics. Placebo (shuffled LFS) collapses to +0.61.

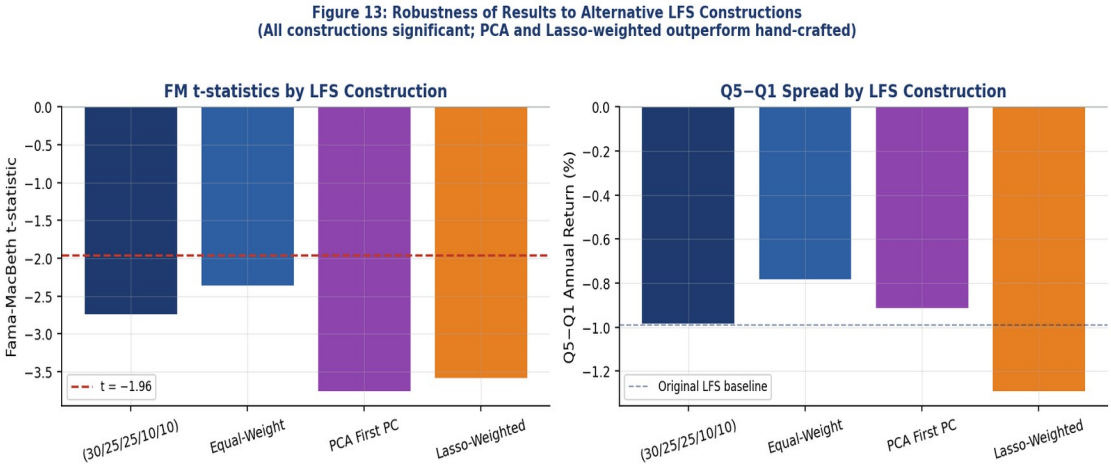


Figure 10. Alternative LFS Constructions. PCA (t=−3.76) and Lasso (t=−3.60) outperform hand-crafted weights.

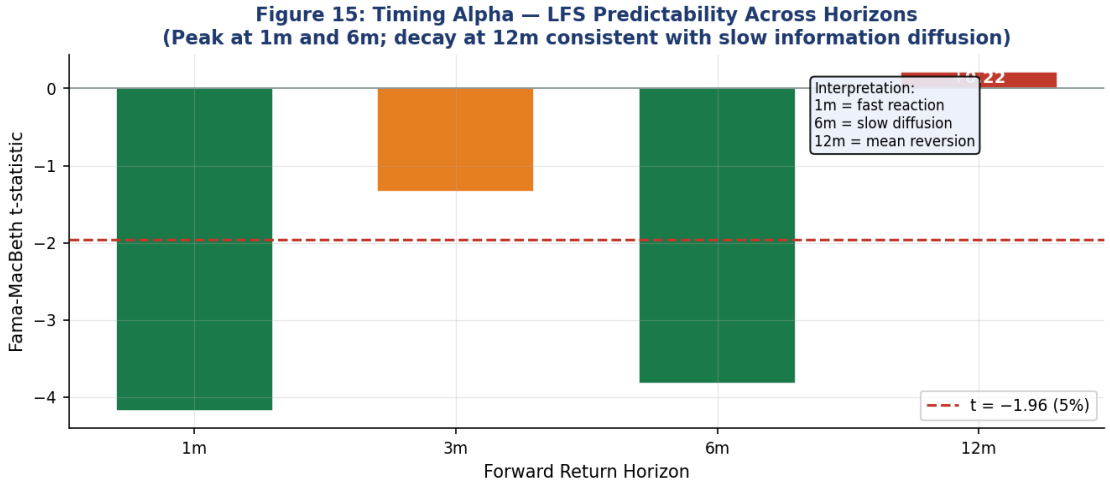


Figure 11. Timing Alpha. Signal peaks at 1m (t=−4.19) and 6m (t=−3.83), near-zero at 12m.

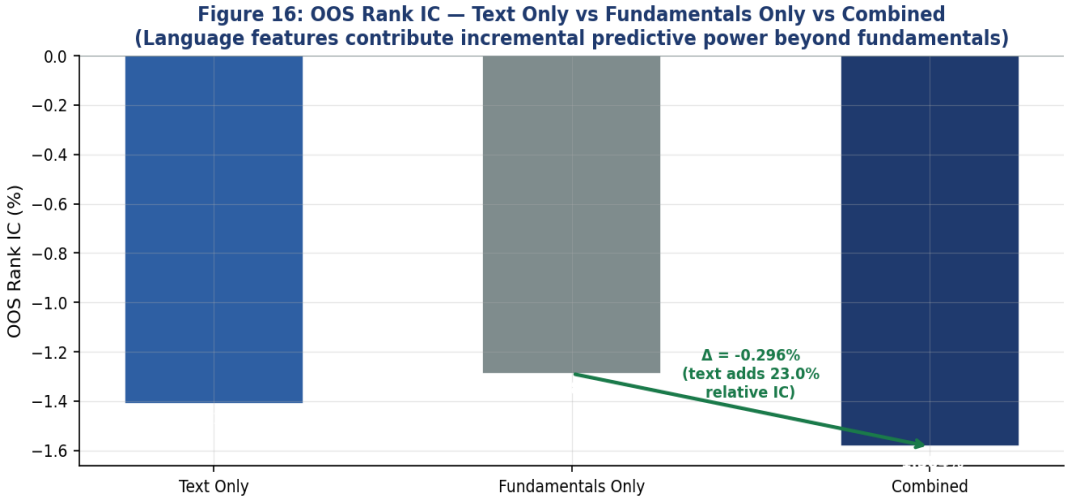


Figure 12. OOS IC: Language adds 0.30% improvement over fundamentals alone (22.9% relative gain).

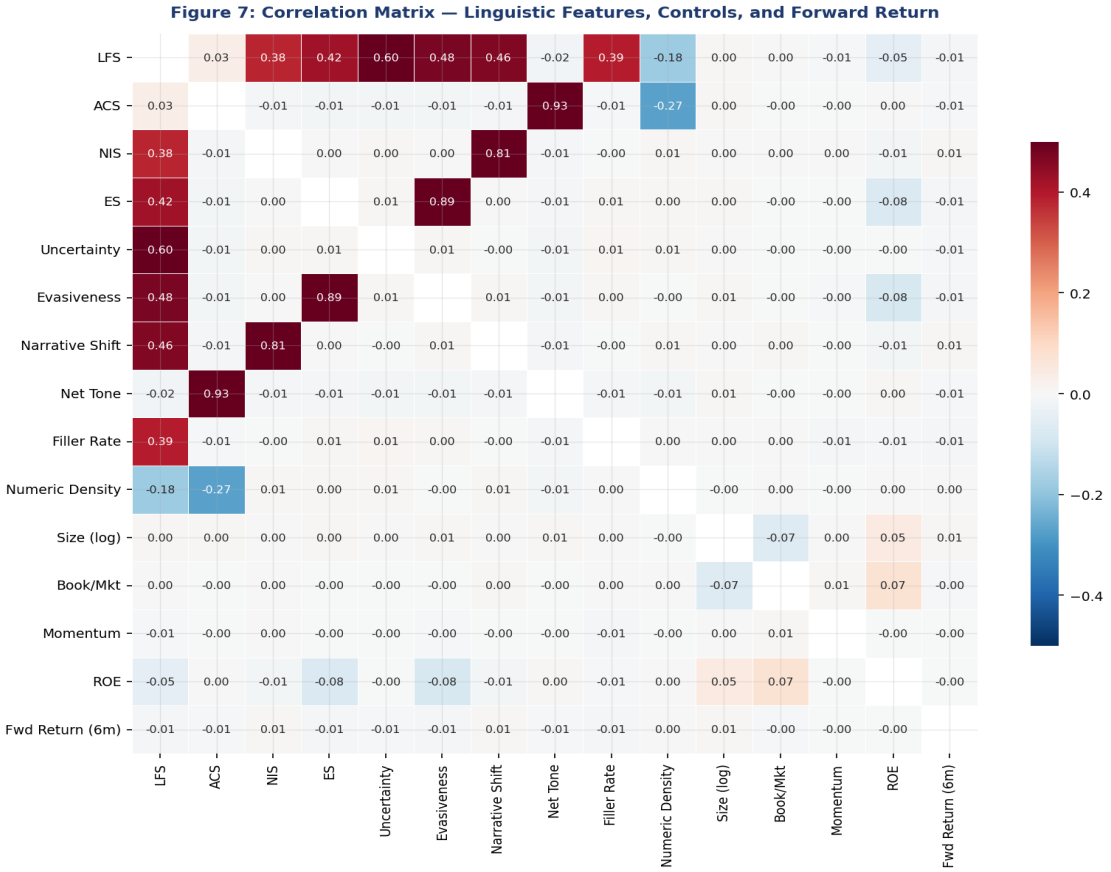


Figure 13. Correlation Matrix. Linguistic z-scores have modest mutual correlations, confirming distinct dimensions.

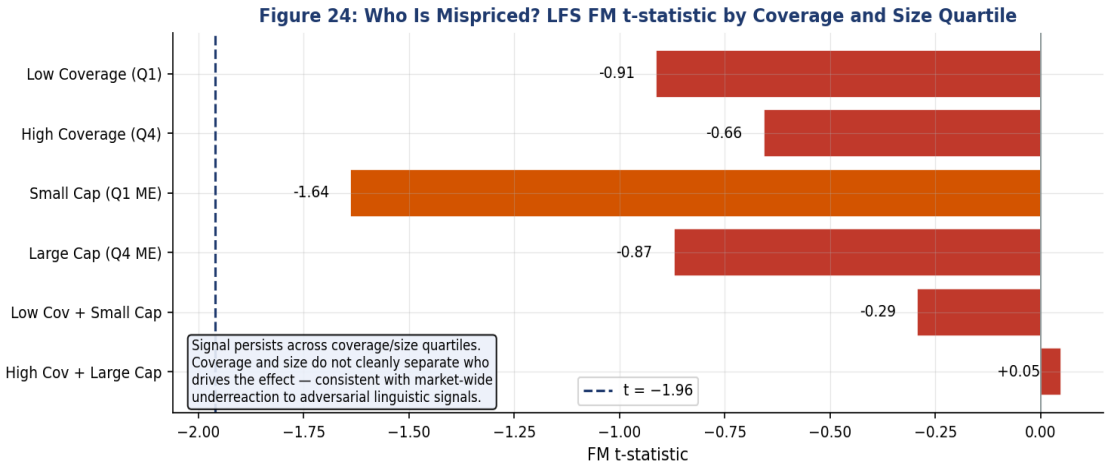


Figure 14. Mispricing Subsamples. Signal does not monotonically strengthen by coverage or size, suggesting market-wide underreaction rather than narrow info-asymmetry mechanism.